

MOHAMMED IQRAM PATEL

AI / SOFTWARE ENGINEER · AGENTIC SYSTEMS & AI PLATFORMS · FULL-STACK ENGINEERING
Mumbai, India | mdiqram@gmail.com | +91 93245 33302 | github.com/Iqram09 | [LinkedIn](#) | [Portfolio](#)

PROFILE

Software engineer specialising in full-stack systems, enterprise API integration, and LLM/agnostic engineering. Built an MCP server for Kong Gateway diagnostics with a deterministic diagnostic engine, structural read-only safety guarantees, a 28-case evaluation harness, and call-level observability. Professional background in enterprise API gateway architecture (Kong, CA Layer7, Apigee) across multiple client projects, spanning requirements through release.

TECHNICAL SKILLS

AI & Agentic Systems: MCP (Model Context Protocol), LLM applications, AI agents, tool-contract design, agent loop design, context management & budgeting, prompt engineering, agentic development workflows

LLM Platform & Ops: Provider abstraction, deterministic evaluation harnesses & scoring, LLM observability, token/cost instrumentation, audit logging, AI guardrails, failure-mode analysis

Languages & Frameworks: TypeScript, JavaScript, Python, Node.js, React, Next.js, React Native, Redux, C++

APIs & Architecture: REST, GraphQL, OAuth 2.0, JWT, Kong API Gateway, CA Layer7, Apigee, Zod schema validation, Postman, service-oriented integration

Cloud & DevOps: Docker, Docker Compose, Git/GitHub, CI/CD, Google Cloud Platform, AWS, Linux

Databases & Testing: SQL Server, MongoDB, Firebase, DynamoDB; Vitest, unit/integration/E2E testing

Data & ML: Pandas, NumPy, Scikit-learn, EDA, Matplotlib, Seaborn

AI ENGINEERING

Kong AI Gateway Diagnostics — TypeScript, MCP SDK v2, Kong 3.9, Docker, Zod, Vitest, Gemini API [repo](#)

- Built an MCP server exposing Kong Gateway diagnostics as 14 read-only, schema-validated tools, letting an LLM agent investigate misrouted services, authentication gaps, and unhealthy upstreams against a live gateway.
- Separated fact from inference by design: a deterministic rule engine (11 pure rules, ~64 ms, no model) produces every finding while the model only sequences tools and writes the explanation — detecting all 10 seeded defects with no false positive on the healthy control.
- Enforced a structural read-only guarantee: no write method exists in the codebase, write tools are gated at registration, the server refuses to start in write mode, and credential values are stripped at five boundaries, each with its own test.
- Wrote a custom agent loop with step limits, identical-call detection, a cumulative context budget, and a halt path that returns an explicit "diagnosis incomplete" rather than a guess; cut context cost to 11% of a raw config dump via entity references, config projection, and model-oriented cursor pagination.
- Built a 28-case evaluation harness across 11 scenarios and 3 difficulty tiers with a deterministic scorer (no LLM judge), false-positive controls, and quota-aware resumable runs — current run passes 28/28 at \$0.0053 per case.
- Instrumented per-call telemetry (correlation IDs, audit log, latency percentiles); 170 automated tests across unit, integration, and end-to-end suites, with 12 observed failure modes documented alongside root cause, mitigation, and regression test.

PROFESSIONAL EXPERIENCE

Associate Consultant · Allied Globetech, Mumbai

May 2025 – Feb 2026

- Designed and integrated enterprise API gateway solutions using Kong, CA Layer7, and Apigee across 3+ client projects, standardising authentication and service integration patterns.
- Built internal React/TypeScript/MongoDB reporting platforms giving consulting teams real-time visibility into project KPIs, replacing manual spreadsheet-based reporting.
- Worked across requirements, implementation, testing, deployment, and release sign-off, collaborating directly with client stakeholders throughout Agile delivery cycles.

Software Developer · Take Solutions, Mumbai

May 2024 – Apr 2025

- Built and maintained full-stack web applications across frontend, backend, API integration, testing, and deployment.
- Designed and consumed REST APIs supporting core application workflows, integrating frontend applications with backend services.
- Participated throughout the delivery lifecycle — sprint planning, automated testing, CI/CD, deployment, and production support — collaborating with cross-functional teams to translate business requirements into maintainable features.

OTHER PROJECTS

E-commerce Dashboard — Next.js, TypeScript, Node.js, MongoDB, TailwindCSS · Product management, order tracking, and admin reporting dashboard.

Food Delivery App — React Native, Redux, Google Maps, Sanity.io, AWS Amplify · Real-time map tracking, cart state management, headless CMS backend.

Streaming Platform — Next.js, Firebase Auth, Stripe, TailwindCSS · Authentication, billing integration, and server-side rendering.

EDUCATION

B.Sc. Information Technology · Mumbai University, 2020 – 2023

Data Science Bootcamp + Virtual Internship · Codebasics, ongoing